



COMPUTATIONAL METHODS IN FINANCE

Computational project 1

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1 Introduction

This project explores numerical methods for option pricing under the Black–Scholes framework. The analysis is developed in two parts, following the structure of the assignment. The first part concerns with the numerical solution of the transformed Black–Scholes equation in the case of a European put option, while the second part addresses the pricing of an American put option, for which the possibility of early exercise leads to a free-boundary problem. The computational setting and model parameters are those specified in the project statement, allowing direct comparison with the results produced in the lecture notes.

For the European option problem, two numerical approaches are considered. The first is the Crank–Nicolson finite-difference method, which provides an implicit discretisation with good stability and accuracy properties. The second is the Method of Lines, in which the spatial variable is discretised first and the resulting system of ordinary differential equations is integrated in time with the classical fourth-order Runge–Kutta method. The purpose of this part of the project is not only to compute the numerical solution, but also to compare the behaviour of these two approaches through the resulting solution surfaces and time profiles.

The second part of the project focuses on the American put option. In this case, the pricing problem is more involved because the option may be exercised before maturity, so the solution must satisfy both the differential equation and the early-exercise constraint. To handle this feature, the numerical procedure combines the Crank–Nicolson discretisation with the Projected Successive Over-relaxation (PSOR) method. This allows the continuation region to be identified numerically and represented graphically.

The main goal of the project is therefore to present the numerical treatment of both the European and American put option problems within the Black–Scholes model, with the formulation of the computational methods, the implementation choices, and the interpretation of the numerical results.

2 Mathematical Model

In the Black–Scholes model, the value $V(S, t)$ of an option depends on the price $S > 0$ of the underlying asset and on time $t \in [0, T]$. A fundamental simplification of the model is that the volatility σ and the interest rate r are assumed to be constant, that is, independent of both time and the asset price. Under the usual no-arbitrage assumptions, the option price satisfies the Black–Scholes equation [2]

$$\frac{\partial V}{\partial t} + \frac{\sigma^2}{2} S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad S > 0, \quad 0 \leq t \leq T.$$

Here, r is the risk-free interest rate and σ is the volatility of the underlying asset.

This is a terminal-value problem, since the payoff is prescribed at maturity $t = T$. For numerical purposes, the unbounded domain is truncated to

$$R_T^V = \{(S, t) \mid 0 < S < S^*, 0 \leq t \leq T\},$$

where $S^* > 0$ is chosen sufficiently large so that the boundary condition at $S = S^*$ approximates the asymptotic behaviour of the solution.

Black–Scholes equation can be obtained from the standard heat equation by a suitable change of variables [2]. Consider the heat equation

$$\frac{\partial y}{\partial \tau} = \frac{\partial^2 y}{\partial x^2}.$$

Introduce the transformation

$$W(x, \tau) = e^{\alpha x + \beta \tau} y(x, \tau).$$

Then

$$\frac{\partial W}{\partial \tau} = e^{\alpha x + \beta \tau} \left(\frac{\partial y}{\partial \tau} + \beta y \right), \quad \frac{\partial W}{\partial x} = e^{\alpha x + \beta \tau} \left(\frac{\partial y}{\partial x} + \alpha y \right),$$

$$\frac{\partial^2 W}{\partial x^2} = e^{\alpha x + \beta \tau} \left(\frac{\partial^2 y}{\partial x^2} + 2\alpha \frac{\partial y}{\partial x} + \alpha^2 y \right).$$

Since y satisfies the heat equation, substituting into the previous expressions gives

$$\frac{\partial W}{\partial \tau} = \frac{\partial^2 W}{\partial x^2} - 2\alpha \frac{\partial W}{\partial x} + (\alpha^2 + \beta)W.$$

Now define

$$x = \log\left(\frac{S}{K}\right), \quad \tau = \frac{\sigma^2}{2}t, \quad U(S, t) = K W(x, \tau),$$

with $S = Ke^x$. By the chain rule,

$$\frac{\partial U}{\partial t} = K \frac{\sigma^2}{2} \frac{\partial W}{\partial \tau}, \quad \frac{\partial U}{\partial S} = K \frac{1}{S} \frac{\partial W}{\partial x}, \quad \frac{\partial^2 U}{\partial S^2} = K \frac{1}{S^2} \left(\frac{\partial^2 W}{\partial x^2} - \frac{\partial W}{\partial x} \right).$$

Substituting these relations into the equation for W , after straightforward algebra, yields

$$\frac{\partial U}{\partial t} = \frac{\sigma^2}{2} S^2 \frac{\partial^2 U}{\partial S^2} + \frac{\sigma^2}{2} (1 - 2\alpha) S \frac{\partial U}{\partial S} + \frac{\sigma^2}{2} (\alpha^2 + \beta) U.$$

To recover the Black–Scholes equation, we choose the coefficients so that

$$\frac{\sigma^2}{2} (1 - 2\alpha) = r, \quad \frac{\sigma^2}{2} (\alpha^2 + \beta) = -r.$$

Equivalently, introducing the notation

$$k = \frac{2r}{\sigma^2},$$

we obtain

$$\alpha = \frac{1 - k}{2}, \quad \beta = -\frac{(k + 1)^2}{4}.$$

With this choice, U satisfies

$$\frac{\partial U}{\partial t} = \frac{\sigma^2}{2} S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S} - rU, \quad (S, t) \in R_T^V.$$

Thus, after the logarithmic change of the spatial variable and the exponential rescaling of the dependent variable, the heat equation is transformed into the Black–Scholes equation [2].

In the present project, however, it is more convenient to replace the terminal-value formulation for V by an initial-value formulation. For that purpose, we reverse time and define

$$U(S, t) = V(S, T - t).$$

Using the chain rule,

$$\frac{\partial U}{\partial t}(S, t) = -\frac{\partial V}{\partial t}(S, T - t), \quad \frac{\partial U}{\partial S}(S, t) = \frac{\partial V}{\partial S}(S, T - t), \quad \frac{\partial^2 U}{\partial S^2}(S, t) = \frac{\partial^2 V}{\partial S^2}(S, T - t).$$

Substituting these relations into the Black–Scholes equation yields the equivalent initial–boundary value problem

$$\frac{\partial U}{\partial t} = \frac{\sigma^2}{2} S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S} - rU, \quad (S, t) \in R_T^V.$$

The terminal condition for V becomes an initial condition for U ,

$$U(S, 0) = V(S, T) = u_0(S), \quad 0 \leq S \leq S^*,$$

and the boundary conditions become

$$U(0, t) = u_a(t), \quad U(S^*, t) = u_b(t), \quad 0 \leq t \leq T,$$

where the functions u_0 , u_a , and u_b depend on the option under consideration.

After solving the numerical problem for $U(S, t)$, the original option value $V(S, t)$ is recovered by reversing time once more, using the relation

$$V(S, t) = U(S, T - t).$$

Therefore, although the Black–Scholes equation may be derived from the heat equation through the previous transformations, the numerical methods in this project are applied to the time-reversed formulation for $U(S, t)$, and the final solution for the option price is obtained by transforming back to the original time variable, as described in the lecture notes [2].

2.1 European put option

For a European put option with strike price K , the payoff at maturity is

$$V(S, T) = \max(K - S, 0).$$

After the change of variables, the initial condition becomes

$$u_0(S) = \max(K - S, 0).$$

The boundary behaviour follows from the limiting properties of the option value. When the asset price approaches zero, the put option remains valuable and its discounted value evolves according to

$$u_a(t) = K e^{-rt}.$$

For sufficiently large asset prices, the put option becomes worthless, which leads to the boundary condition

$$u_b(t) = 0.$$

These conditions define a well-posed initial–boundary value problem that can be solved numerically using finite-difference methods or semi-discrete approaches.

2.2 American put option

In the case of an American put option, early exercise is allowed before maturity. As a result, the option value must satisfy an inequality constraint in addition to the differential equation. The transformed solution must remain above the payoff function,

$$U(S, t) \geq \max(K - S, 0).$$

Furthermore, whenever the option value coincides with the payoff, the differential operator satisfies a complementary condition. This leads to a linear complementarity formulation in which the numerical solution must simultaneously satisfy the discretised differential equation and the early-exercise constraint.

The boundary conditions for the American put remain consistent with those of the European case,

$$U(0, t) = K, \quad U(S^*, t) = 0,$$

while the initial condition is again given by the payoff function,

$$U(S, 0) = \max(K - S, 0).$$

The presence of the inequality constraint introduces a free-boundary structure, separating the continuation region, where the differential equation holds, from the exercise region, where the option value equals the payoff. Numerical methods must therefore be adapted to enforce this constraint at each time step.

3 Numerical Methods

The transformed Black–Scholes problem was solved numerically using finite-difference discretisation in the spatial variable combined with appropriate time-integration schemes. A uniform grid was introduced in both variables. The spatial domain $[0, S^*]$ was divided into N_S subintervals of size $h_S = S^*/N_S$, while the time interval $[0, T]$ was divided into N_t steps of size $h_t = T/N_t$.

Let $S_i = ih_S$ for $i = 0, \dots, N_S$ and $t_n = nh_t$ for $n = 0, \dots, N_t$. The numerical approximation of $U(S_i, t_n)$ is denoted by U_i^n . Spatial derivatives were approximated using second-order central finite-differences. In particular, the first and second derivatives with respect to S were approximated by

$$\frac{\partial U}{\partial S}(S_i, t_n) \approx \frac{U_{i+1}^n - U_{i-1}^n}{2h_S}, \quad \frac{\partial^2 U}{\partial S^2}(S_i, t_n) \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{h_S^2}.$$

The central finite-difference approximations used for the first and second spatial derivatives are second-order accurate in space. More precisely, the local truncation error associated with these approximations is of order $O(h_S^2)$, provided that the solution is sufficiently smooth. These approximations lead to tridiagonal matrix

structures that can be solved efficiently at each time step.

3.1 Crank–Nicolson method

For the European option problem, the time discretisation was performed using the Crank–Nicolson method. This scheme is obtained by averaging the spatial operator between two consecutive time levels. As a result, the method is second-order accurate in both space and time. The spatial accuracy follows from the central finite-difference approximations, while the temporal accuracy results from the symmetric treatment of the time derivative.

At each time step, the method produces a linear system of the form

$$AU^{n+1} = BU^n + b^n,$$

where A and B are tridiagonal matrices determined by the model parameters and the discretisation steps. Boundary values are imposed explicitly at each time level and contribute to the right-hand side vector b^n . The resulting linear systems were solved using sparse matrix techniques, exploiting the tridiagonal structure of the coefficient matrices.

The stability of the Crank–Nicolson method can be analysed using Von Neumann stability analysis. In this approach, the numerical solution is expressed as a Fourier mode and the corresponding amplification factor is obtained. The analysis shows that the amplification factor satisfies $|\eta| \leq 1$, independently of the discretisation parameters h_S and h_t . Therefore, the Crank–Nicolson method is unconditionally stable, meaning that stability is preserved for arbitrary time-step sizes. This property makes the method particularly suitable for solving parabolic partial differential equations such as the Black–Scholes equation.

3.2 Method of Lines with Runge–Kutta time integration

As an alternative approach, the Method of Lines was used by first discretizing only the spatial derivatives. This leads to a system of ordinary differential equations in time of the form

$$\frac{dW}{dt} = AW + b(t),$$

where $W(t)$ represents the vector of interior solution values and A is a tridiagonal matrix resulting from the spatial discretisation.

The resulting semi-discrete system was integrated in time using the classical fourth-order Runge–Kutta method. This method provides fourth-order accuracy in time, with local truncation error of order $O(h_t^4)$.

Since the classical Runge–Kutta scheme is explicit, the method is conditionally stable. The time-step size must satisfy a restriction related to the eigenvalues of the matrix A resulting from the spatial discretisation. For diffusion-type problems such as the Black–Scholes equation, this typically requires smaller time steps than those permitted by implicit methods.

3.3 American option and PSOR method

For the American put option, the presence of the early-exercise constraint leads to a linear complementarity problem at each time step. After discretisation using the Crank–Nicolson method, the numerical solution must satisfy both the discretised equation and the inequality constraint associated with the payoff.

To solve this problem, the Projected Successive Over-relaxation (PSOR) method was employed. At each time step, the linear system arising from the Crank–Nicolson discretisation was solved iteratively. After each iteration, the solution was projected onto the admissible region defined by the payoff condition, ensuring that

$$U_i^{n+1} \geq \max(K - S_i, 0).$$

The iteration continued until the difference between successive approximations fell below a prescribed tolerance or until a maximum number of iterations was reached. A relaxation parameter was used to accelerate convergence of the iterative procedure.

3.4 Comparison of accuracy and stability properties

Both numerical approaches employed in this work share the same spatial discretisation based on central finite-differences. Therefore, the spatial approximation error for both methods is of order $O(h_S^2)$.

The difference between the two methods arises primarily from the temporal discretisation. The Crank–Nicolson method is second-order accurate in time, while the classical fourth-order Runge–Kutta scheme used in the Method of Lines approach provides temporal accuracy of order $O(h_t^4)$.

The Crank–Nicolson method is unconditionally stable, as established through Von Neumann stability analysis in the lecture notes, independently of the choice of the spatial and temporal step sizes.

In contrast, the Method of Lines combined with an explicit Runge–Kutta time integrator is conditionally

stable. After spatial discretisation, the problem reduces to a system of ordinary differential equations, whose stability depends on the time step relative to the eigenvalues of the semi-discrete operator. Consequently, sufficiently small time steps are required to obtain stable solutions.

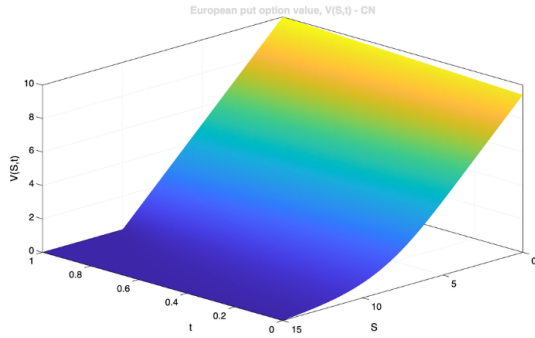
Table 1: Theoretical comparison between numerical methods.

Method	Space Order	Time Order	Stability
Crank–Nicolson	$O(h_S^2)$	$O(h_t^2)$	Unconditional
MOL + RK4	$O(h_S^2)$	$O(h_t^4)$	Conditional

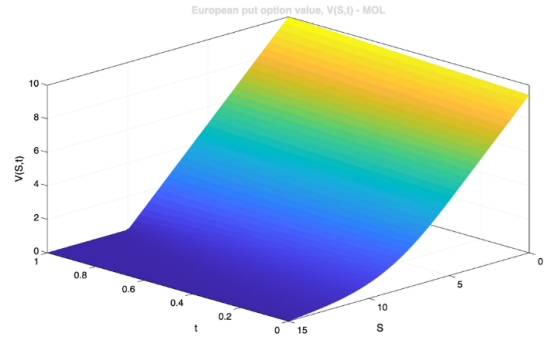
4 European Put Option Results

The numerical solution of the European put option problem was computed using two distinct approaches: the Crank–Nicolson finite-difference method and the Method of Lines combined with fourth-order Runge–Kutta time integration. The purpose of this comparison is to assess the consistency of the numerical results obtained from these two formulations and to verify their agreement with the qualitative properties expected from the Black–Scholes model.

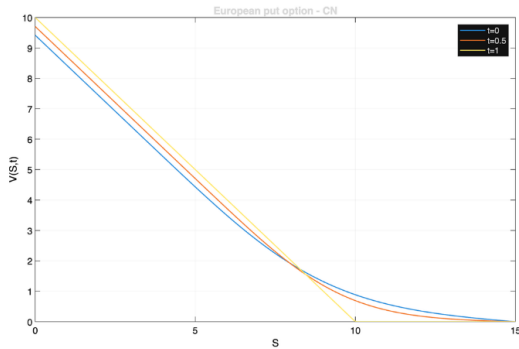
The computed solution surfaces are shown in Figure 1. The upper row displays the option value surface obtained using the Crank–Nicolson method, while the lower row shows the corresponding surface obtained using the Method of Lines formulation. In both cases, the solution evolves smoothly from the initial payoff condition and maintains the expected monotonic dependence on the asset price variable.



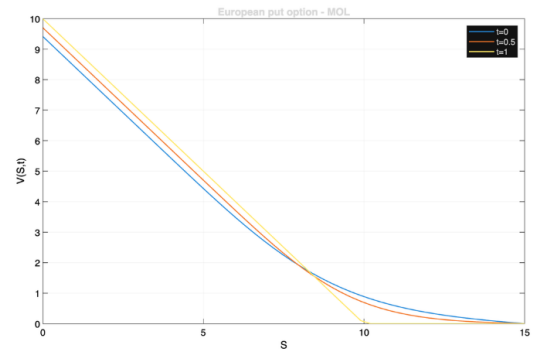
(a) European put surface obtained using the Crank–Nicolson method.



(b) European put surface obtained using the Method of Lines.



(c) European put profiles at selected times using Crank–Nicolson.



(d) European put profiles at selected times using the Method of Lines.

Figure 1: European put option numerical results obtained using two numerical approaches. The upper row shows the solution surfaces obtained using Crank–Nicolson and the Method of Lines, respectively. The lower row shows time profiles of the solution at selected time levels, confirming the consistency between both methods.

The comparison between the two methods shows very similar numerical behaviour across the domain. No visible numerical oscillations were observed, indicating that the selected discretisation parameters were adequate to ensure stability.

5 American Put Option Results

The valuation of the American put option introduces an additional layer of complexity relative to the European case due to the possibility of early exercise. In contrast to the European option, whose value is determined solely by the solution of the Black–Scholes partial differential equation, the American option must satisfy an inequality constraint ensuring that the option value remains at least equal to its intrinsic value at all times. This

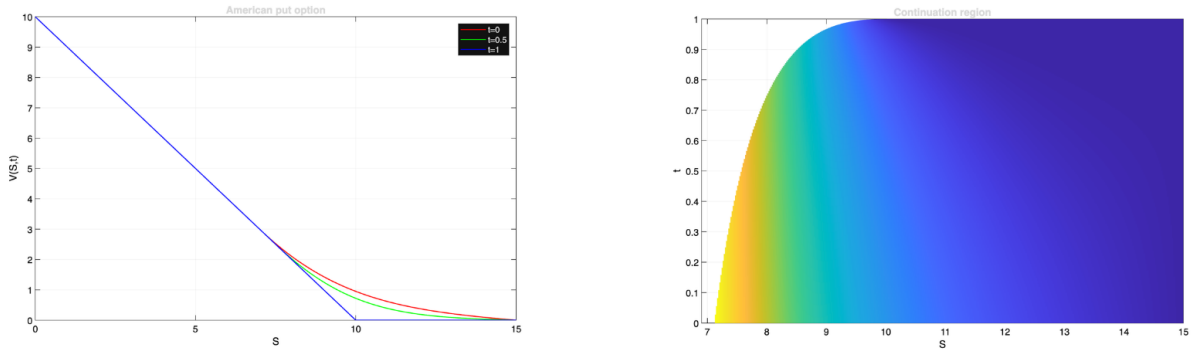
transforms the numerical problem into a linear complementarity problem, which was solved using the Projected Successive Over-Relaxation (PSOR) method combined with the Crank–Nicolson time-stepping scheme.

At each time level, the discretised Crank–Nicolson system produces a tridiagonal linear system similar to that encountered in the European case. However, instead of solving this system directly, the PSOR algorithm was applied iteratively. After each iteration step, the computed values were projected onto the admissible region defined by the early exercise constraint. Specifically, the option value at each spatial node was enforced to remain greater than or equal to the intrinsic payoff function. This projection step ensures convergence toward a solution that satisfies both the differential equation and the early exercise condition.

Figure 2a presents the American put option profiles evaluated at selected time levels. These profiles illustrate the characteristic deviation from the European solution, particularly in regions where early exercise becomes optimal. Near maturity, the solution closely follows the intrinsic payoff, while for earlier times the option value remains strictly above the payoff in regions where holding the option retains time value.

A key feature of American option pricing is the emergence of the continuation region, defined as the set of points in the (S, t) domain where it is optimal to retain the option rather than exercise it immediately. This region is bounded by the early exercise frontier, which evolves backward in time from maturity. The numerical solution obtained using the PSOR method allows the identification of this continuation region by comparing the computed option value with the intrinsic payoff.

Figure 2b shows the continuation region obtained from the numerical solution. The region corresponds to areas where the option value exceeds the payoff, indicating that continuation is optimal. Conversely, regions where the option value equals the payoff correspond to immediate exercise. The shape of this region reflects the economic intuition that early exercise becomes more favorable when the underlying asset price is sufficiently low relative to the strike price.



(a) American put option profiles obtained using Crank–Nicolson and PSOR.

(b) Continuation region and early exercise boundary.

Figure 2: American put option numerical results illustrating both the time profiles of the solution and the continuation region separating the exercise and holding domains.

The numerical results confirm that the American put option value is consistently greater than or equal to the corresponding European option value across the computational domain. This difference becomes more pronounced in regions where early exercise is advantageous, particularly for low values of the underlying asset price. The PSOR method demonstrated stable convergence behaviour across time steps, provided that an appropriate relaxation parameter was selected. In practice, convergence was achieved within a moderate number of iterations per time step, highlighting the computational efficiency of the method for this class of problems.

From a financial perspective, the continuation region provides important insight into the optimal exercise policy. The numerical solution reveals that early exercise is primarily concentrated in regions where the option is deep in-the-money, reflecting the diminishing value of waiting when the payoff is already substantial. As time to maturity increases, the continuation region expands, reflecting the increasing importance of time value in the option price.

Overall, the combination of the Crank–Nicolson discretisation with the PSOR iterative method proved effective for solving the American option pricing problem. The results demonstrate that the numerical scheme successfully captures both the early exercise constraint and the associated free boundary behaviour, producing stable and economically consistent solutions across the computational domain.

6 Discussion and Conclusions

The numerical results obtained in this project confirm that the finite-difference methods are an effective approach for pricing options within the Black–Scholes framework. Both European and American put options were successfully computed, producing solutions consistent with the expected qualitative behaviour.

For the European put option, the Crank–Nicolson method provided a smooth and stable numerical solution,

with the computed values converging towards the terminal payoff as maturity approached, as expected. The resulting linear systems were efficiently solved using tridiagonal matrices, for the sake of efficiency.

For the American put option, the presence of the early exercise feature required the enforcement of an inequality constraint. The Projected Successive Over-Relaxation (PSOR) method proved effective in maintaining the complementarity condition and computing the option value at each time step. The resulting solution allowed the identification of the continuation region and the associated free boundary.

The computed continuation region confirmed the expected financial behaviour: early exercise occurs when the option is sufficiently in the money, while continuation becomes more likely for larger maturities due to the time value of the option.

The combination of the Crank–Nicolson discretisation and the PSOR method provides a reliable numerical framework for solving both European and American option pricing problems, being accurate in capturing the early exercise feature and the evolution of the option value over time.

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